

Introduction

In order to help in conservation efforts through reforestation, LRI decided to create a map combining criteria from two studies: First, the Important plant areas map created by Bou Dagher-Kharrat et al. (2018) . Second, a Master thesis by Bercachy (2017) from the Lebanese University that produced reforestation suitability maps for 27 native tree species (*Abies cilicica*, *Acer monspessulanum*, *Acer obtusifolium*, *Acer hyrcanum tauricum*, *Amygdalus orientalis*, *Arbutus andrachne*, *Cedrus libani*, *Ceratonia siliqua*, *Cercis siliquastrum*, *Crataegus azarolus*, *Cupressus sempervirens*, *Fraxinus ornus*, *Juniperus drupacea*, *Juniperus excelsa*, *Laurus nobilis*, *Malus trilobata*, *Pinus pinea*, *Pyrus syriaca*, *Quercus calliprinos*, *Quercus cedrorum*, *Quercus cerris*, *Quercus infectoria*, *Quercus ithaburensis*, *Quercus kotschyana*, *Quercus look*, *Sorbus torminalis*, and *Styrax officinalis*) using a Species Distribution Model (SDM) based on MaxEnt. Bercachy's study used multiple environmental factors such as Digital Elevation Model (DEM), Integrated Moisture Index (IMI), Potential Direct Incident Radiation (PDIR), Emberger Quotient (EQ), Normalized Difference Vegetation Index (NDVI), Cloud Coverage (CC), Distance from the Sea (DFS), Minimal Temperature of the Coldest Month (MinT), and Cumulative Annual Precipitation (P). In our study, we merge these two works to provide a detailed spatial analysis that includes the integration of richness and endemism data from the Bou Dagher-Kharrat grid with the reforestation suitability maps (derived from ASC files showing species distribution probabilities above 0.5, converted to polygons, unioned, and merged) from Bercachy's work. The combined dataset was then used to calculate average species occurrence values per grid cell, with a subsequent correlation analysis ($r = 0.4$) between species count and richness confirming a moderate positive relationship between the two approaches.

Methodology

First, the grid created by Bou Dagher-Kharrat et al. (2018) had a resolution of 3x3 km which provides data on species richness (the number of species present in each grid) and endemism. The map was used as a spatial framework as it provides mapping and visualization of data with good accuracy and fits with the geomorphological properties of Lebanon.

In Bercachy's study, species distribution maps were generated for 27 native tree species (listed above) using MaxEnt. The input data for these models included presence points for each species and several key environmental factors (DEM, IMI, PDIR, EQ, NDVI, CC, DFS,

MinT, and P) selected from a thorough literature review for their known influence on species distribution.

The map was created using ArcGIS following these Steps: Using Raster Calculator multiplied by 10, we got data that we transformed to integers using int tool. Integers with the threshold 5 and above present in the value field representing the probability of species occurrence were extracted and turned to polygons. We did this step for all the 27 species. The extraction process was repeated twice to ensure the validation of extracted data. Afterwards the polygons were merged in a single map of all the species. This map was Unioned with the original IPA to link the values to the grid. This map was then dissolved using.

Next, Using arcpy 2.7 on ArcGIS 10.7.1 the original feature class containing species fields and an "ID_1" identifier was copied to a temporary dataset to preserve the source data. In this temporary copy, new fields (appended with "_NZ") were created for each of the 27 species. An update cursor was used to convert all zero values to NULL, ensuring that only valid occurrence data influenced the calculations.

Following this, the dataset was dissolved by the "ID_1" field, aggregating the data and calculating the mean occurrence for each species in every grid cell. A new field, "Sp_occurrence," was added to store species with an average occurrence value of 5 or above. The resulting map generated from the script did not contain all the fields present in the IPA shapefile. To ensure completeness, I merged it with the IPA, creating a new feature class within a geodatabase to prevent any truncation and preserve the full dataset. Finally, I added a count field to quantify the number of species present in *Sp_occurrence*, which was then used to analyze the correlation between species count and species richness in the original IPA.

Results & Discussion

The outcome is the original IPA shapefile converted into a feature class within a geodatabase with two new added field:

- The “Sp_occurrence” field containing the botanical name of each of the 27 species and the probability of its occurrence for each grid cell.
- The “count” field of the number of species in each cell of the grid.

A correlation study was conducted in Excel to assess the relationship between species richness from the IPA grid and the count field. The analysis resulted in a Pearson correlation coefficient of 0.4, indicating a moderate positive correlation between the two datasets.

Merging these two approaches allowed us to create dependable outcomes. The Bou Dagher-Kharrat grid provided a robust framework based on observed species richness and endemism, while Bercachy's maps contributed species distribution models based on an extensive literature review of environmental factors that most affect habitat suitability. Applying a 0.5 probability threshold ensured that only high-likelihood occurrence areas were considered. The moderate correlation between species counts and richness ($r = 0.4$) validates the integrated approach. A moderate correlation is due to a key difference between the two data sets. While the IPA grid represents present species distributions, the MaxEnt models predict potential habitats. This explains why some high-suitability areas do not align perfectly with recorded richness values. Additional factors such as habitat fragmentation, human disturbance, and local extinction may also influence the correlation. Despite these differences, the relationship between the two datasets supports the relevance of combining current richness data with habitat suitability models for conservation planning.

References

- Bercachy, C. (2017). *LEBANON REFORESTATION SUITABILITY MAP FOR SELECTED TREE SPECIES* [Master of Sciences]. Lebanese University.
- Bou Dagher-Kharrat, M., El Zein, H., & Rouhan, G. (2018). Setting conservation priorities for Lebanese flora—Identification of important plant areas. *Journal for Nature Conservation*, 43, 85–94. <https://doi.org/10.1016/j.jnc.2017.11.004>